GURU GOBIND SINGH POLYTECHNIC, NASIK
COMPUTER ENGINEERING DEPARTMENT

PRACTICAL EXAM QUESTIONS

Subject: Mobile Application Development(22619)

1. **Write a program demonstrate Linear Layout with vertical and horizontal orientation.**

XML

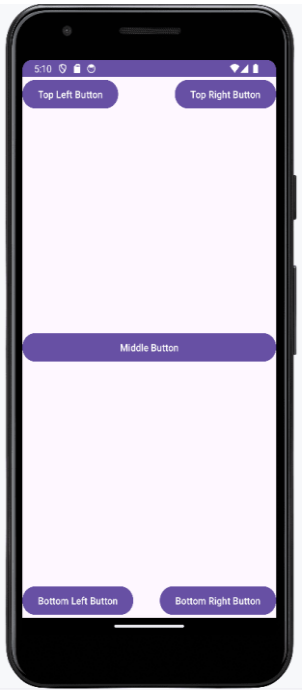
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <!-- Vertical Layout Components -->
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Vertical Layout"
        android:textSize="20sp"
        android:layout_marginBottom="10dp" />

    <Button
        android:id="@+id/btn1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Button 1" />

    <Button
        android:id="@+id/btn2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Button 2"
        android:layout_marginTop="10dp" />

    <!-- Horizontal Layout -->
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:orientation="horizontal"
        android:text="horizontal layout"
        android:textSize="20sp"

        ></TextView>
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal"
        android:layout_marginTop="20dp">

        <Button
```

```
        android:id="@+id/btn3"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Left" />

    <Button
        android:id="@+id/btn4"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Right"
        android:layout_marginStart="10dp" />

  </LinearLayout>

</LinearLayout>
```

JAVA

```
package com.example.myapplication;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

2. **Write a program to produce following output using Relative Layout:**



XML

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="16dp">

    <!-- Top Left Button -->
    <Button
        android:id="@+id/btnTopLeft"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Top Left Button"
        android:layout_alignParentTop="true"
        android:layout_alignParentStart="true" />

    <!-- Top Right Button -->
    <Button
        android:id="@+id/btnTopRight"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Top Right Button"
        android:layout_alignParentTop="true"
        android:layout_alignParentEnd="true" />

    <!-- Middle Button -->
    <Button
        android:id="@+id/btnMiddle"
        android:layout_width="match_parent"
```

```
        android:layout_height="wrap_content"
        android:text="Middle Button"
        android:layout_centerInParent="true" />

    <!-- Bottom Left Button -->
    <Button
        android:id="@+id/btnBottomLeft"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Bottom Left Button"
        android:layout_alignParentBottom="true"
        android:layout_alignParentStart="true" />

    <!-- Bottom Right Button -->
    <Button
        android:id="@+id/btnBottomRight"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Bottom Right Button"
        android:layout_alignParentBottom="true"
        android:layout_alignParentEnd="true" />

</RelativeLayout>
```

3. **Write a program to produce following output using Table Layout:**

XML

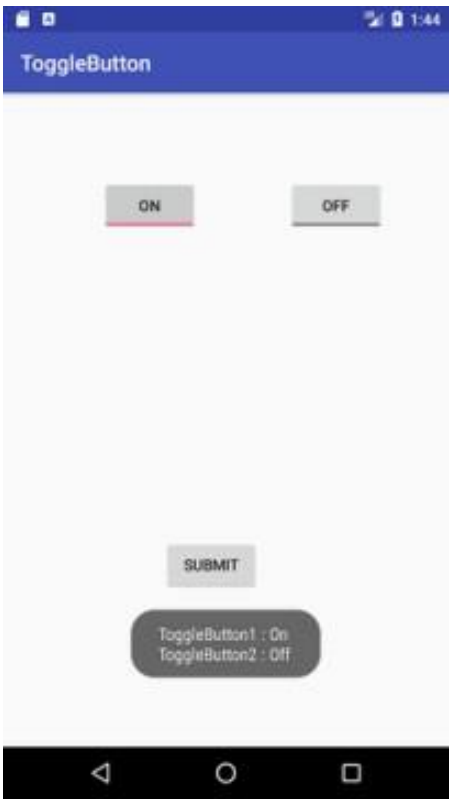
```
<?xml version="1.0" encoding="utf-8"?>
<TableLayout xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:stretchColumns="*"
   android:padding="16dp">

   <!-- Heading -->
   <TableRow>
      <TextView
         android:layout_span="3"
         android:text="TableUI"
         android:textSize="24sp"
         android:textStyle="bold"
         android:gravity="center"
         android:padding="10dp" />
   </TableRow>

   <!-- Header Row -->
   <TableRow>
      <TextView
         android:text="Student Name"
         android:textStyle="bold"
         android:padding="8dp" />

      <TextView
         android:text="Course"
         android:textStyle="bold"
         android:padding="8dp" />

      <TextView
         android:text="Score"
         android:textStyle="bold"
         android:padding="8dp" />
   </TableRow>

   <!-- Row 1 -->
   <TableRow>
      <TextView
         android:text="Rajeevan"
         android:padding="8dp" />

      <TextView
         android:text="Ancient Mythology"
         android:padding="8dp" />

      <TextView
```

```
        android:text="100"
        android:padding="8dp" />
    </TableRow>

    <!-- Row 2 -->
    <TableRow>
      <TextView
        android:text="Krishna"
        android:padding="8dp" />

      <TextView
        android:text="Politics"
        android:padding="8dp" />

      <TextView
        android:text="98"
        android:padding="8dp" />
    </TableRow>

    <!-- Row 3 -->
    <TableRow>
      <TextView
        android:text="Chanakya"
        android:padding="8dp" />

      <TextView
        android:text="Economics"
        android:padding="8dp" />

      <TextView
        android:text="98"
        android:padding="8dp" />
    </TableRow>

</TableLayout>
```

4. **Write a program to Frame Layout to display Text over Image.**

```
<?xml version="1.0" encoding="utf-8"?>
<FrameLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <!-- Background Image -->
    <ImageView
        android:id="@+id/imageView"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:src="@drawable/sample_image"
        android:scaleType="centerCrop" />

    <!-- Text Over Image -->
    <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Welcome to MAD"
        android:textSize="24sp"
        android:textStyle="bold"
        android:textColor="#FFFFFF"
        android:layout_gravity="center" />

</FrameLayout>
```

5. **Write a program to produce following output using Toggle Button:**



XML

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <!-- First Toggle Button -->
    <ToggleButton
        android:id="@+id/toggleButton1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textOn="ON"
        android:textOff="OFF" />

    <!-- Second Toggle Button -->
    <ToggleButton
        android:id="@+id/toggleButton2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textOn="ON"
        android:textOff="OFF"
        android:layout_marginTop="20dp" />

    <!-- Submit Button -->
    <Button
        android:id="@+id/button"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="SUBMIT"
        android:layout_marginTop="40dp" />
```

```
    <!-- Output Text -->
    <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="18sp"
        android:padding="20dp"
        android:layout_marginTop="20dp" />

</LinearLayout>
```

JAVA

```
package com.example.togglebuttondemo;

import android.os.Bundle;
import android.widget.Button;
import android.widget.TextView;
import android.widget.ToggleButton;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    ToggleButton toggleButton1, toggleButton2;
    Button button;
    TextView textView;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        toggleButton1 = findViewById(R.id.toggleButton1);
```

```
        toggleButton2 = findViewById(R.id.toggleButton2);

        button = findViewById(R.id.button);

        textView = findViewById(R.id.textView);


        button.setOnClickListener(v -> {

            String state1 = toggleButton1.isChecked() ? "On" : "Off";

            String state2 = toggleButton2.isChecked() ? "On" : "Off";


            textView.setText("ToggleButton1 : " + state1 +

                    "\nToggleButton2 : " + state2);

        });

    }

}
```

6. Write a program to design simple login Screen. Check User name and password and if it is correct, display a toast message, “Login Success”, otherwise, “Login Failed”. Assume suitable GUI.

XML

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Login Screen"
        android:textSize="24sp"
        android:textStyle="bold"
        android:layout_marginBottom="20dp"/>

    <EditText
        android:id="@+id/etUsername"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Username" />

    <EditText
        android:id="@+id/etPassword"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Password"
        android:inputType="textPassword"
        android:layout_marginTop="10dp" />
```

```
        <Button
            android:id="@+id/btnLogin"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="LOGIN"
            android:layout_marginTop="20dp" />

    </LinearLayout>
```

JAVA

```
package com.example.logindemo;

import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText etUsername, etPassword;
    Button btnLogin;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        etUsername = findViewById(R.id.etUsername);
        etPassword = findViewById(R.id.etPassword);
        btnLogin = findViewById(R.id.btnLogin);

        btnLogin.setOnClickListener(v -> {
            String username = etUsername.getText().toString();
            String password = etPassword.getText().toString();

            if (username.equals("admin") && password.equals("1234")) {
                Toast.makeText(MainActivity.this,
                        "Login Success",
                        Toast.LENGTH_SHORT).show();
            } else {
                Toast.makeText(MainActivity.this,
                        "Login Failed",
                        Toast.LENGTH_SHORT).show();
            }
```

```
        });
    }
}
```

7. **Write a program to demonstrate Date Picker. Display Selected date using Toast. Assume suitable GUI.**

XML

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"

    android:layout_width="match_parent"

    android:layout_height="match_parent"

    android:orientation="vertical"

    android:gravity="center"

    android:padding="20dp">



    <DatePicker

        android:id="@+id/datePicker"

        android:layout_width="wrap_content"

        android:layout_height="wrap_content" />



    <Button

        android:id="@+id/btnShowDate"

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:text="SHOW DATE"

        android:layout_marginTop="20dp" />
```

```
</LinearLayout>
```

JAVA

```
package com.example.datepickerdemo;


import android.os.Bundle;

import android.widget.Button;

import android.widget.DatePicker;

import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;


public class MainActivity extends AppCompatActivity {


    DatePicker datePicker;

    Button btnShowDate;


    @Override

    protected void onCreate(Bundle savedInstanceState) {

        super.onCreate(savedInstanceState);

        setContentView(R.layout.activity_main);


        datePicker = findViewById(R.id.datePicker);

        btnShowDate = findViewById(R.id.btnShowDate);
```

```
        btnShowDate.setOnClickListener(v -> {

            int day = datePicker.getDayOfMonth();

            int month = datePicker.getMonth() + 1;

            int year = datePicker.getYear();



            String selectedDate = day + "/" + month + "/" + year;



            Toast.makeText(MainActivity.this,

                    "Selected Date: " + selectedDate,

                    Toast.LENGTH_SHORT).show();

        });

    }

}
```

8. **Write a program to demonstrate Time Picker. Display Selected Time using Toast. Assume suitable GUI.**

XML

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"

    android:layout_width="match_parent"

    android:layout_height="match_parent"

    android:orientation="vertical"

    android:gravity="center"
```

```
    android:padding="20dp">



    <TimePicker

        android:id="@+id/timePicker"

        android:layout_width="wrap_content"

        android:layout_height="wrap_content" />



    <Button

        android:id="@+id/btnShowTime"

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:text="SHOW TIME"

        android:layout_marginTop="20dp" />



</LinearLayout>
```

JAVA

```
package com.example.timepickerdemo;
```

```
import android.os.Bundle;

import android.widget.Button;

import android.widget.TimePicker;

import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;



public class MainActivity extends AppCompatActivity {



    TimePicker timePicker;

    Button btnShowTime;



    @Override

    protected void onCreate(Bundle savedInstanceState) {

        super.onCreate(savedInstanceState);

        setContentView(R.layout.activity_main);



        timePicker = findViewById(R.id.timePicker);

        btnShowTime = findViewById(R.id.btnShowTime);
```

```
        btnShowTime.setOnClickListener(v -> {

            int hour = timePicker.getHour();

            int minute = timePicker.getMinute();



            String selectedTime = hour + ":" + minute;



            Toast.makeText(MainActivity.this,

                    "Selected Time: " + selectedTime,

                    Toast.LENGTH_SHORT).show();

        });

    }

}
```

9. **Write a program to demonstrate Implicit and Explicit Intent. Assume suitable GUI.**

JAVA

MAIN ACCTIVITY

```
package com.example.intentdemo;

import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.widget.Button;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    Button btnExplicit, btnImplicit;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        btnExplicit = findViewById(R.id.btnExplicit);
        btnImplicit = findViewById(R.id.btnImplicit);

        // Explicit Intent
        btnExplicit.setOnClickListener(v -> {
            Intent intent = new Intent(MainActivity.this, SecondActivity.class);
            startActivity(intent);
        });

        // Implicit Intent
        btnImplicit.setOnClickListener(v -> {
            Intent intent = new Intent(Intent.ACTION_VIEW);
            intent.setData(Uri.parse("https://www.google.com"));
            startActivity(intent);
        });
    }
}
```

SECOND ACTIVITY

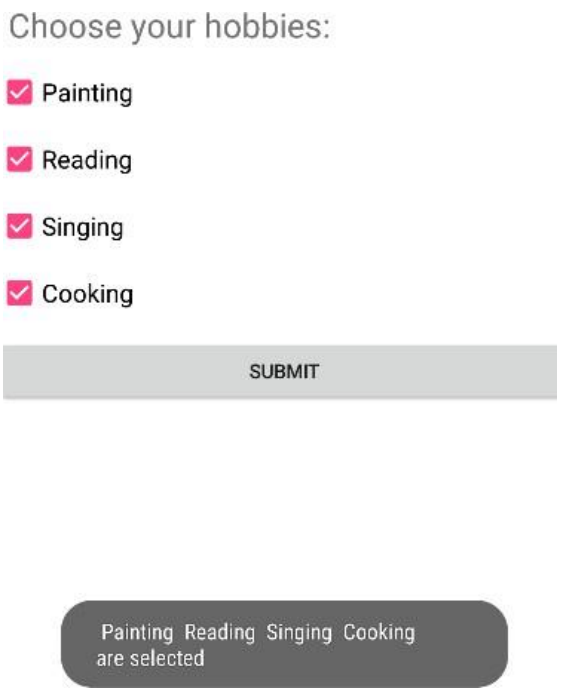
```
package com.example.intentdemo;

import android.os.Bundle;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

public class SecondActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);

        TextView textView = new TextView(this);
        textView.setText("This is Second Activity");
```

```
        textView.setTextSize(24);

        setContentView(textView);
    }
}
```

XML

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <Button
        android:id="@+id/btnExplicit"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Explicit Intent" />

    <Button
        android:id="@+id/btnImplicit"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Implicit Intent"
        android:layout_marginTop="20dp" />

</LinearLayout>
```

10. **Write a program to produce following output:**



XML

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Choose your hobbies:"
        android:textSize="22sp"
        android:textStyle="bold"
        android:layout_marginBottom="20dp"/>

    <CheckBox
        android:id="@+id/cbPainting"
        android:layout_width="wrap_content"
```

```
        android:layout_height="wrap_content"
        android:text="Painting" />

    <CheckBox
        android:id="@+id/cbReading"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Reading" />

    <CheckBox
        android:id="@+id/cbSinging"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Singing" />

    <CheckBox
        android:id="@+id/cbCooking"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Cooking" />

    <Button
        android:id="@+id/btnSubmit"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="SUBMIT"
        android:layout_marginTop="20dp" />

</LinearLayout>
```

JAVA

```
package com.example.checkboxdemo;

import android.os.Bundle;
```

```
import android.widget.Button;
import android.widget.CheckBox;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    CheckBox cbPainting, cbReading, cbSinging, cbCooking;
    Button btnSubmit;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        cbPainting = findViewById(R.id.cbPainting);
        cbReading = findViewById(R.id.cbReading);
        cbSinging = findViewById(R.id.cbSinging);
        cbCooking = findViewById(R.id.cbCooking);
        btnSubmit = findViewById(R.id.btnSubmit);

        btnSubmit.setOnClickListener(v -> {
            String selected = "";

            if (cbPainting.isChecked())
                selected += "Painting ";
            if (cbReading.isChecked())
                selected += "Reading ";
            if (cbSinging.isChecked())
                selected += "Singing ";
            if (cbCooking.isChecked())
                selected += "Cooking ";

            Toast.makeText(MainActivity.this,
```

```
                selected + "are selected",
                Toast.LENGTH_SHORT).show();
        });
    }
}
```

11. Write a program to produce following output:

XML

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Select Gender"
        android:textSize="22sp"
        android:textStyle="bold"
        android:layout_marginBottom="40dp"/>

    <RadioGroup
        android:id="@+id/radioGroupGender"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:orientation="horizontal"
        android:layout_marginBottom="50dp">
```

```
        <RadioButton
            android:id="@+id/radioMale"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Male"
            android:textSize="18sp"
            android:checked="true"
            android:layout_marginEnd="30dp"/>

        <RadioButton
            android:id="@+id/radioFemale"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Female"
            android:textSize="18sp"/>
    </RadioGroup>

    <Button
        android:id="@+id/btnCheckGender"
        android:layout_width="200dp"
        android:layout_height="wrap_content"
        android:text="Check Gender"
        android:layout_marginBottom="60dp"/>

    <Button
        android:id="@+id/btnResult"
        android:layout_width="180dp"
        android:layout_height="wrap_content"
        android:text="Male"/>

</LinearLayout>
```

JAVA

```
package com.example.gendertest;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.RadioButton;
import android.widget.RadioGroup;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    private RadioGroup radioGroupGender;
    private Button btnCheckGender;
    private Button btnResult;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // Find views
```

```
        radioGroupGender = findViewById(R.id.radioGroupGender);
        btnCheckGender = findViewById(R.id.btnCheckGender);
        btnResult = findViewById(R.id.btnResult);

        // Default value
        btnResult.setText("Male");

        // Button click listener
        btnCheckGender.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                int selectedId = radioGroupGender.getCheckedRadioButtonId();

                if (selectedId != -1) {
                    RadioButton selectedRadio = findViewById(selectedId);
                    String gender = selectedRadio.getText().toString();

                    btnResult.setText(gender);

                    Toast.makeText(MainActivity.this,
                            "Selected: " + gender,
                            Toast.LENGTH_SHORT).show();
                } else {
                    Toast.makeText(MainActivity.this,
                            "Please select a gender",
                            Toast.LENGTH_SHORT).show();
                }
            }
        });
    }
}
```

12. **Write a program to produce following output using ListView:**

XML

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical">

    <!-- ListView -->
    <ListView
        android:id="@+id/listView"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:divider="#CCCCCC"
        android:dividerHeight="1dp"/>

    </LinearLayout>
```

JAVA

```
package com.example.countryapp;

import android.os.Bundle;
import android.view.View;
import android.widget.AdapterView;
import android.widget.ArrayAdapter;
```

```
import android.widget.ListView;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    ListView listView;
    String[] countries = {
            "India",
            "China",
            "australia",
            "Portugle",
            "America",
            "NewZealand"
    };

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        listView = findViewById(R.id.listView);

        // Simple ArrayAdapter with default layout
        ArrayAdapter<String> adapter = new ArrayAdapter<>(
                this,
                android.R.layout.simple_list_item_1,
                countries
        );

        listView.setAdapter(adapter);

        // Optional: Show toast when item is clicked
        listView.setOnItemClickListener(new AdapterView.OnItemClickListener() {
            @Override
            public void onItemClick(AdapterView<?> parent, View view, int position, long id) {
                String selectedCountry = countries[position];
                Toast.makeText(MainActivity.this,
                        "You selected: " + selectedCountry,
                        Toast.LENGTH_SHORT).show();
            }
        });
```

```
    }
    }
```

13. **Write a program to produce following output using GridView:**

XML

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical">

    <!-- Blue Title Bar -->
    <TextView
        android:layout_width="match_parent"
        android:layout_height="50dp"
        android:background="#1976D2"
        android:text="GridViewExample"
        android:textColor="#FFFFFF"
        android:textSize="18sp"
        android:gravity="center_vertical"
        android:paddingLeft="16dp"/>

    <!-- GridView -->
    <GridView
        android:id="@+id/gridView"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:numColumns="5"
        android:verticalSpacing="10dp"
```

```
        android:horizontalSpacing="10dp"
        android:padding="10dp"
        android:stretchMode="columnWidth"/>

</LinearLayout>
```

JAVA

```
package com.example.gridviewexample;

import android.os.Bundle;
import android.widget.ArrayAdapter;
import android.widget.GridView;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        GridView gridView = findViewById(R.id.gridView);

        // Array of alphabets A to Z
        String[] letters = {
            "A", "B", "C", "D", "E",
            "F", "G", "H", "I", "J",
            "K", "L", "M", "N", "O",
            "P", "Q", "R", "S", "T",
            "U", "V", "W", "X", "Y",
            "Z"
        };

        // Simple ArrayAdapter
        ArrayAdapter<String> adapter = new ArrayAdapter<>(
                this,
                android.R.layout.simple_list_item_1,
                letters
        );

        gridView.setAdapter(adapter);
```

15. Write a program to convert length to feet. Assume suitable GUI.

XML

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Length: Meters to Feet"
        android:textSize="24sp"
        android:textStyle="bold"
        android:layout_marginBottom="20dp" />

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal"
        android:gravity="center_vertical">

        <EditText
            android:id="@+id/etMeters"
```

```
        android:layout_width="0dp"
        android:layout_weight="1"
        android:layout_height="wrap_content"
        android:hint="100.0"
        android:inputType="numberDecimal"
        android:background="@android:drawable/editbox_background_normal"
        android:padding="10dp" />

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Meters"
        android:textSize="18sp"
        android:layout_marginStart="10dp" />
  </LinearLayout>

  <Button
    android:id="@+id/btnConvert"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="CONVERT TO FEET"
    android:backgroundTint="#3498db"
    android:textColor="#FFFFFF"
    android:layout_marginTop="20dp" />

  <TextView
    android:id="@+id/tvResult"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Result: 0.00 Feet"
    android:textSize="20sp"
    android:layout_marginTop="20dp" />

</LinearLayout>
```

JAVA

```
package com.example.simpleconverter;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;
```

```
import java.util.Locale;

public class MainActivity extends AppCompatActivity {

    // Declare UI components
    private EditText etMeters;
    private Button btnConvert;
    private TextView tvResult;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // Link components to the XML IDs
        etMeters = findViewById(R.id.etMeters);
        btnConvert = findViewById(R.id.btnConvert);
        tvResult = findViewById(R.id.tvResult);

        // Set the click listener for the button
        btnConvert.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                convertMetersToFeet();
            }
        });
    }

    private void convertMetersToFeet() {
        String input = etMeters.getText().toString();

        if (!input.isEmpty()) {
            try {
                double meters = Double.parseDouble(input);

                // Formula: 1 Meter = 3.28084 Feet
                double feet = meters * 3.28084;
```

```
            // Format to 2 decimal places to match your image
            String resultText = String.format(Locale.getDefault(), "Result: %.2f Feet", feet);
            tvResult.setText(resultText);

        } catch (NumberFormatException e) {
            tvResult.setText("Invalid Input");
        }
    } else {
        tvResult.setText("Please enter a value");
    }
  }
}
```

16. Write a program to convert INR to USD. Assume suitable GUI.

XML

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="INR to USD Converter"
        android:textSize="24sp"
        android:textStyle="bold"
        android:layout_marginBottom="20dp"/>

    <EditText
        android:id="@+id/etInr"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Amount in INR"
        android:inputType="numberDecimal"/>

    <Button
        android:id="@+id/btnConvert"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="CONVERT"
        android:layout_marginTop="20dp"/>

    <TextView
        android:id="@+id/tvResult"
        android:layout_width="wrap_content"
```

```
        android:layout_height="wrap_content"
        android:textSize="20sp"
        android:layout_marginTop="20dp"/>

</LinearLayout>
```

JAVA

```
package com.example.currencyconverter;

import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText etInr;
    Button btnConvert;
    TextView tvResult;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        etInr = findViewById(R.id.etInr);
        btnConvert = findViewById(R.id.btnConvert);
        tvResult = findViewById(R.id.tvResult);

        btnConvert.setOnClickListener(v -> {
            double inr = Double.parseDouble(etInr.getText().toString());
            double usd = inr / 83;

            tvResult.setText("USD = $" + usd);
        });
    }
}
```

17. Write a program to implement Splash Screen.

XML
MAIN

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
```

```
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">

    <TextView
        android:id="@+id/tvHome"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Home Screen"
        android:textSize="28sp"
        android:textStyle="bold" />

</LinearLayout>
```

ACTIVITY SPLASH

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:background="#6200EE">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Welcome to My App"
        android:textColor="#FFFFFF"
        android:textSize="30sp"
        android:textStyle="bold"/>

</LinearLayout>
```

JAVA
MAINACTIVITY

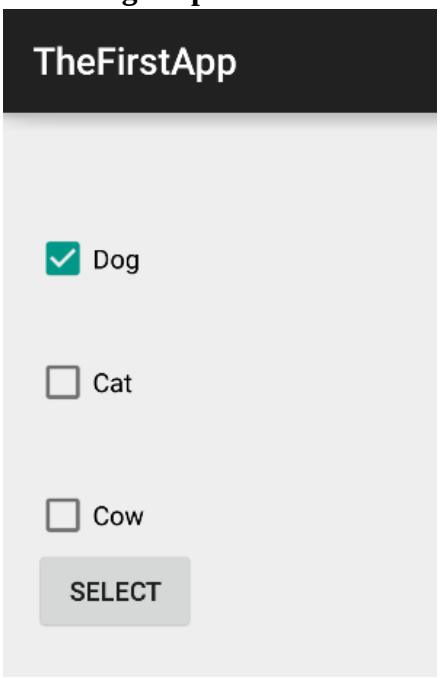
```
package com.example.splashdemo;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

SPLASH ACTIVITY

```
package com.example.splashdemo;

import android.content.Intent;
import android.os.Bundle;
import android.os.Handler;
import androidx.appcompat.app.AppCompatActivity;
```

```
public class SplashActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_splash);

        new Handler().postDelayed(() -> {
            Intent intent = new Intent(SplashActivity.this, MainActivity.class);
            startActivity(intent);
            finish();
        }, 3000); // 3 seconds
    }
}
```

18. Write a program to produce the following output. On button click, display selected options:



XML

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="30dp">

    <CheckBox
        android:id="@+id/cbDog"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Dog"
        android:textSize="20sp"
        android:layout_marginBottom="20dp"/>

    <CheckBox
        android:id="@+id/cbCat"
        android:layout_width="wrap_content"
```

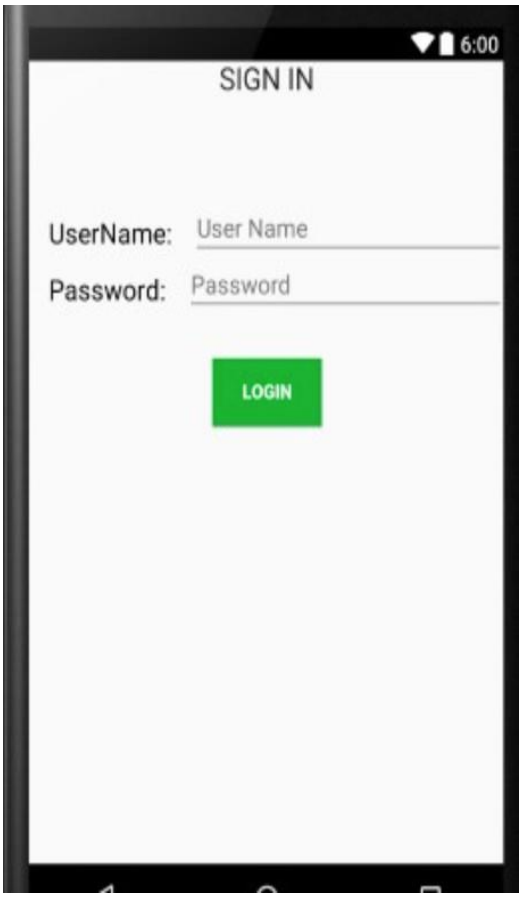
```
        android:layout_height="wrap_content"
        android:text="Cat"
        android:textSize="20sp"
        android:layout_marginBottom="20dp"/>

    <CheckBox
        android:id="@+id/cbCow"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Cow"
        android:textSize="20sp"
        android:layout_marginBottom="20dp"/>

    <Button
        android:id="@+id/btnSelect"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="SELECT"
        android:padding="15dp"/>

</LinearLayout>
```

JAVA

```
package com.example.thefirstapp;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.CheckBox;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    private CheckBox cbDog, cbCat, cbCow;
    private Button btnSelect;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // Initialize UI components
        cbDog = findViewById(R.id.cbDog);
        cbCat = findViewById(R.id.cbCat);
        cbCow = findViewById(R.id.cbCow);
        btnSelect = findViewById(R.id.btnSelect);

        btnSelect.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                StringBuilder result = new StringBuilder("Selected: ");

                if (cbDog.isChecked()) {
```

```
                result.append("\nDog");
            }
            if (cbCat.isChecked()) {
                result.append("\nCat");
            }
            if (cbCow.isChecked()) {
                result.append("\nCow");
            }

            if (result.toString().equals("Selected: ")) {
                Toast.makeText(MainActivity.this, "Nothing selected",
    Toast.LENGTH_SHORT).show();
            } else {
                Toast.makeText(MainActivity.this, result.toString(),
    Toast.LENGTH_LONG).show();
            }
        }
      });
    }
}
```

19. Write a program to produce following:



XML

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp"
    android:gravity="center_horizontal">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="SIGN IN"
        android:textSize="22sp"
        android:textColor="#333333"
        android:layout_marginBottom="40dp" />

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal"
        android:layout_marginBottom="15dp">

        <TextView
            android:layout_width="100dp"
```

```
        android:layout_height="wrap_content"
        android:text="UserName:"
        android:textSize="18sp"
        android:textColor="#000000" />

    <EditText
        android:id="@+id/etUser"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="User Name"
        android:inputType="textPersonName" />
  </LinearLayout>

  <LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:layout_marginBottom="30dp">

    <TextView
        android:layout_width="100dp"
        android:layout_height="wrap_content"
        android:text="Password:"
        android:textSize="18sp"
        android:textColor="#000000" />

    <EditText
        android:id="@+id/etPass"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Password"
        android:inputType="textPassword" />
  </LinearLayout>

  <Button
    android:id="@+id/btnLogin"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="LOGIN"
    android:backgroundTint="#27ae60"
    android:textColor="#FFFFFF"
    android:paddingLeft="30dp"
    android:paddingRight="30dp" />

</LinearLayout>
```

JAVA

```
package com.example.signinapp;

import android.os.Bundle;
```

```
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    private EditText etUser, etPass;
    private Button btnLogin;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // Initialize components
        etUser = findViewById(R.id.etUser);
        etPass = findViewById(R.id.etPass);
        btnLogin = findViewById(R.id.btnLogin);

        btnLogin.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String username = etUser.getText().toString();
                String password = etPass.getText().toString();

                if (username.isEmpty() || password.isEmpty()) {
                    Toast.makeText(MainActivity.this, "Please enter all details",
Toast.LENGTH_SHORT).show();
                } else {
                    // Typical validation logic would go here
                    String message = "Logging in: " + username;
                    Toast.makeText(MainActivity.this, message, Toast.LENGTH_SHORT).show();
                }
            }
        });
    }
}
```

20. Write a program to produce the following using ImageButton. Onclick of Button, show Toast Message, which button was clicked.

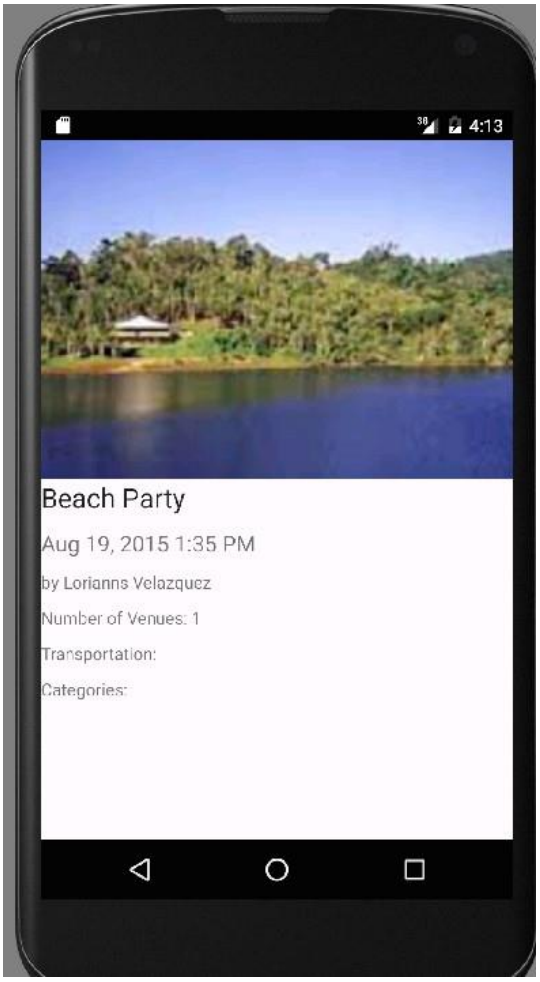
XML

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <ImageButton
        android:id="@+id/btnHome"
        android:layout_width="150dp"
        android:layout_height="150dp"
        android:src="@drawable/home_icon"
        android:scaleType="fitCenter"
        android:background="?android:attr/selectableItemBackground"
        android:layout_marginBottom="30dp" />

    <ImageButton
        android:id="@+id/btnYoutube"
        android:layout_width="150dp"
        android:layout_height="150dp"
        android:src="@drawable/youtube_icon"
        android:scaleType="fitCenter"
        android:background="?android:attr/selectableItemBackground"
        android:layout_marginBottom="20dp" />

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="YouTube Button"
        android:background="@android:color/darker_gray"
        android:textColor="@android:color/white"
        android:padding="10dp" />

</LinearLayout>
```

JAVA

```java
package com.example.imagebuttonapp;

import android.os.Bundle;
import android.view.View;
import android.widget.ImageButton;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    private ImageButton btnHome, btnYoutube;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // Initialize ImageButtons
        btnHome = findViewById(R.id.btnHome);
        btnYoutube = findViewById(R.id.btnYoutube);

        // Set listener for Home Button
        btnHome.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                Toast.makeText(MainActivity.this, "Home Button Clicked", Toast.LENGTH_SHORT).show();
            }
        });

        // Set listener for YouTube Button
        btnYoutube.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                Toast.makeText(MainActivity.this, "YouTube Button Clicked", Toast.LENGTH_SHORT).show();
            }
        });
    }
}
```

21. Write a program to design following output using ImageView and TextView:



XML

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:background="#FFFFFF">

   <LinearLayout
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:orientation="vertical">

      <ImageView
         android:id="@+id/ivHeader"
         android:layout_width="match_parent"
         android:layout_height="250dp"
         android:scaleType="centerCrop"
         android:src="@drawable/beach_image" />

      <LinearLayout
         android:layout_width="match_parent"
         android:layout_height="wrap_content"
         android:orientation="vertical"
         android:padding="16dp">

         <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
```

```
            android:text="Beach Party"
            android:textSize="24sp"
            android:textColor="#000000"
            android:layout_marginBottom="8dp" />

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Aug 19, 2015 1:35 PM"
            android:textSize="16sp"
            android:textColor="#757575"
            android:layout_marginBottom="12dp" />

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="by Lorianns Velazquez"
            android:textSize="14sp"
            android:textColor="#9E9E9E"
            android:layout_marginBottom="8dp" />

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Number of Venues: 1"
            android:textSize="14sp"
            android:textColor="#9E9E9E"
            android:layout_marginBottom="8dp" />

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Transportation:"
            android:textSize="14sp"
            android:textColor="#9E9E9E"
            android:layout_marginBottom="8dp" />

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Categories:"
            android:textSize="14sp"
            android:textColor="#9E9E9E" />

    </LinearLayout>
  </LinearLayout>
</ScrollView>
```

JAVA

```
package com.example.eventapp;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
```

```
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        // Hiding the action bar to match the full-screen look of the image
        if (getSupportActionBar() != null) {
            getSupportActionBar().hide();
        }
        setContentView(R.layout.activity_main);
    }
}
```

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